Emerging Technology Plan: Quantum Computing

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Overview of Quantum Computing

- Uses qubits for parallel computation
- Supports complex simulations
- Transforming finance, healthcare
- Enables new cryptography models
- High performance potential



Key Research Findings

- Superior performance for simulations
- Scalability challenges with qubits
- Security implications for encryption
- Long-term high ROI potential
- Industry growth projected



Industry Applications

- Financial risk modeling
- Portfolio optimization
- Drug discovery
- Logistics optimization
- Cryptography transition



Case Study: JPMorgan & IBM Quantum

- Partnership began 2020
- Focus on financial simulations
- Hybrid-cloud model access
- Quantum-safe cryptography prep
- Improved modeling outcomes



Risk, Security & Compliance

- Threatens current encryption
- Requires quantum-safe methods
- High implementation cost
- Talent skill gaps
- Regulatory concerns



Scalability Potential

- Cloud quantum access growing
- Qubit stability improving
- New architectures emerging
- Projected multi-industry adoption
- High long-term performance



Sustainability & Maintenance Plan

- Invest in talent development
- Use hybrid-cloud quantum model
- Adopt NIST post-quantum standards
- Continuous performance testing
- Vendor partnerships essential



Key Lessons & Recommendations

- Adopt pilot-first strategy
- Plan crypto transition early
- Measure KPIs regularly
- Invest in R&D ecosystem
- Gradually scale workloads



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